TransKAN: Transfer Learning and Kolmogorov Arnold Networks for Gunshot Wound Classification

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Abstract. *Determining wound orientation (entry or exit) and estimating medico-legal shooting distance (MLSD) remain highly complex forensic challenges. These tasks rely on specialized forensics experts and remain highly sensitive to crime scene illumination and imaging conditions. Despite the recent advancements in machine learning (ML) techniques, particularly deep learning (DL) and its applications in computer vision (CV), its applications in these forensics tasks remain under-explored. Recent progress in DL has led to the development of automated classifiers that achieve high performance by processing raw gunshot wound imagery. However, despite these efforts, techniques reported in the literature for gunshot wound classification (GSWC) perform below expectations when compared to other CV tasks, such as diabetic retinopathy screening, medical image classification, biometric identification [Freitas et al. 2023b, Fioravanti et al. 2024], and many others. To overcome these challenges, we introduce TransKAN, a novel hybrid architecture that leverages transfer learning and Kolmogorov-Arnold networks (KANs). Specifically, TransKAN leverages a pre-trained ResNet152 network to extract high-level spatial features, substituting the conventional dense classifier with a KAN to capture intricate non-linear relationships more efficiently. In order to validate TransKAN, we present Gunshot Wound Image Database (GuWID), a comprehensive database created using real photographic records of homicide and suicide cases and examined by crime scene forensic experts from incidents attended by Federal District Civil Police (PCDF). Images within the GuWID dataset were collected between January 2012 and March 2022, comprising a total of 2,551 images (1,883 entry and 668 exit wounds). Categorized by MLSD, the dataset includes 1,609 distant, 232 close-range, and 42 contact shots. Experimental results show that TransKAN demonstrate achieves state-of-the-art (SOTA) performance on the GuWID dataset.*

1. Introduction

Handguns are a primary instrument of global homicide, accounting for three-quarters of lethal acts in the Americas in 2021 [UNODC 2023], with Brazil alone exceeding 47,000 annual firearm deaths [Wisevoter 2024]. Forensic experts must therefore differentiate entry from exit wounds and classify the range at which they were inflicted [Kleypas and Badiye 2024, Shrestha et al. 2024], based on shape, dimensions, and border characteristics [Denton et al. 2006] that can be obscured, missing, or mixed. Projectile velocity,